
COURSE	CECS 378	TERM	Su26
EXAM	Exam 1	POINTS	50

Exam 1 — Introduction to Computer Security

🔔 **VERIFY YOUR NAME AND STUDENT ID BELOW ARE CORRECT, THEN ADD TODAY'S DATE.**

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STUDENT ID#: 040100101

DATE: _____

🔔 **Directions.** Write on the exam paper. You may use one 3"×5" card of notes (both sides, handwritten only). No electronic devices of any kind are allowed on your desk during the exam.

SECTION 1 — Multiple Choice (4 pts each, 20 pts total)

- (4 pts) Which of the following best describes a **symmetric-key cipher**?
 - The sender encrypts with a public key; the receiver decrypts with a private key.
 - The same secret key is used for both encryption and decryption.
 - No key is required; security relies entirely on algorithm secrecy.
 - A digital signature is produced as a side effect of encryption.
- (4 pts) A cryptographic hash function is said to be **collision resistant** if:
 - Given a hash value, it is computationally infeasible to find any input that produces it.
 - The hash output is exactly the same length as the input.
 - It is computationally infeasible to find two distinct inputs that produce the same hash output.
 - The hash function runs in constant time regardless of input length.
- (4 pts) In RSA, a message is encrypted for a recipient by using:
 - The sender's private key.
 - A shared session key negotiated via Diffie-Hellman.
 - The recipient's private key.
 - The recipient's public key.
- (4 pts) A classic **stack buffer overflow** typically overwrites which memory region to redirect control flow?
 - The heap allocation header.
 - The process's environment variables.
 - The saved return address on the stack frame.
 - The kernel's system-call dispatch table.
- (4 pts) The **principle of least privilege** states that:
 - A subject should be granted the maximum privileges needed for any possible future task.
 - All users must share a single privileged account to simplify auditing.
 - Privileges are determined by a mandatory label assigned at object creation time.

- (d) A subject should be granted only the minimum privileges necessary to perform its current task.

SECTION 2 — Short Answer (10 pts each, 30 pts total)

6. (10 pts) Briefly explain **why symmetric ciphers are generally used to encrypt bulk data** in practice, while asymmetric ciphers are typically used only for key exchange or digital signatures. What fundamental trade-off motivates this hybrid design?

7. (10 pts) Name and briefly describe **two** OS/compiler-level mitigations for stack buffer overflow attacks. For each mitigation, also state one way an attacker might attempt to circumvent it.

8. (10 pts) Compare **Discretionary Access Control (DAC)** and **Mandatory Access Control (MAC)**.

For each model: (i) who controls access decisions, and (ii) give one concrete example of a system or OS feature that uses it.

End of exam. Total: 50 points.

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